M3 — Sweeps as one program, optimisers, gradient noise (N1, O3, N2)

jax-experiments writeup · M3

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Status: N1 and O3 done; N2 not started ·
Hardware: Kaggle T4 ×2 (sm_75, one job per GPU), JAX 0.11.1; CPU (4 cores) for the equivalence tests ·
Data: N1: `data/n1-summary.json` and `data/n1/` (one JSON and log per job); O3: `data/o3-summary.json` and `data/o3/`; ml-ladder's references from its `docs/writeups/data/c-gpu-lr.json` and `c-gpu-lr2-1024.json` ·
Reproduce: `uv run python tools/kaggle/push_r2.py <preset> --push` for `n1-calibrate`, `n1`, `n1-fork` and `n1-fork-torch`, then `uv run python experiments/n1_vmap_sweep/analyze.py`; O3: presets `o3-stage1` to `o3-stage4`, then `uv run python experiments/o3_optimizers/analyze.py`. CPU tests: `uv run pytest tests/test_n1_sweep.py tests/test_o3_optimizers.py`

N1 — Hyperparameter sweeps as one program

Question

ml-ladder's Part C §7 tuned the 4M transformer's learning rate with 18 separate runs. Can one vmapped program train all of them at once, and how much faster is it?

The idea

A sweep is K copies of the same computation. In JAX, training is a pure function `step(params, opt_state, batch, lr)`. Stack K copies of every argument on a leading axis, and `jit(vmap(step))` trains all of them in one compiled program. The learning rate is just another vmapped argument.

So is everything else that differs between the runs:

- each copy's loss scale;
- its skipped steps;
- its batch stream: ml-ladder's runs of one seed share batches, so copies of one seed share one draw per step.

Two questions, kept apart:

- **Is it the same experiment?** Each copy of (lr, seed) starts from ml-ladder's initial weights for that seed and sees that seed's batches, so it can be paired with ml-ladder's run.
- **Is it faster?** Total throughput of K copies in one program, against one copy, which is the rate of K sequential runs.

The math

`vmap` turns every operation on a $[\ldots]$-shaped array into one on a $[K, \ldots]$-shaped array:

- a matmul xW becomes a batched matmul $x_k W_k$;
- per-copy scalars (the loss, the gradient norm used for clipping, the loss-scale decision) become $[K]$-vectors.

Nothing couples the copies, so copy k computes exactly what its sequential run would, up to the order of floating-point reductions.

The speedup has a simple ceiling. If one copy already keeps the GPU's arithmetic units busy, K copies take K times as long. vmap can only win back idle time: kernel-launch overhead, small matmuls that underfill the device, Python between steps.

What we measure and why

- **Equivalence (CPU).** Two seeds × two learning rates, vmapped against four sequential runs. Checked in fp32 and fp16, for a GRU and a transformer.
- **The sweep (T4).** ml-ladder's §7 grid: peak learning rates {3e-4, 6e-4, 1e-3, 2e-3, 4e-3, 8e-3} × seeds 0–2, for the 4M transformer at T=1024. Each run is 3,000 steps of 16 × 1024 bytes, then per-position bits/byte on 262,144 validation bytes. It ran as six programs of three copies.
- **Memory.** XLA's compiled temp memory for K = 1 to 18 copies, with and without per-block remat.
- **Throughput.** Steady tokens/s for K = 1, 2, 4, 8 (and up to 32 for small models), over 200 steps each. Measured on ml-ladder's 4M model and on two much smaller transformers (width 128 and 64, depth 2) at the same 16 × 1024 tokens per step.

What I built

- **`jaxexp.sweep.train_many`:**
  - stacks K copies of parameters, optimiser state and loss scale;
  - runs `jit(vmap(make_step(...)))`;
  - follows each copy's own schedule;
  - draws each unique batch stream once per step.

  It returns one `TrainResult` per copy.
- **`tests/test_n1_sweep.py`:** vmapped copies equal sequential runs.
- **`experiments/n1_vmap_sweep/`:**
  - `run.py`, with `memory`, `throughput` and `sweep` studies;
  - `analyze.py`.
- **The Kaggle runner** now accepts jobs for any experiment script, and gives JAX 92% of the card instead of 75%.

Results

On CPU, vmap computes what the sequential runs compute.

	loss histories	all parameters (relative)	outputs on held-out data
fp32	within 10^{-6}	within 10^{-6}	within 10^{-6}
fp16	within 3×10^{-3}	within 3×10^{-3}	within 3×10^{-3}

Measured, the fp32 parameter differences are 9×10^{-8} to 3×10^{-7}.

One tensor needed explaining: the attention key bias. Per tensor it differs by up to 5% in fp16. Its true gradient is exactly zero, because adding a constant vector to every key shifts all of a query's logits equally, and the softmax ignores that. So its gradient is pure round-off, which Adam scales up to full-size steps in either program. It has no effect on the outputs.

On the T4, the sweep reproduces ml-ladder's curve.

peak LR	ml-ladder bits/byte	JAX, vmapped (3 per program)	largest per-seed difference
3e-4	3.0279 (3.0259–3.0322)	3.0272 (3.0261–3.0312)	0.0010

peak LR	ml-ladder bits/byte	JAX, vmapped (3 per program)	largest per-seed difference
6e-4	2.4353 (2.4041–2.5574)	2.4358 (2.4002–2.5548)	0.0040
1e-3	2.0441 (2.0123–2.2154)	2.0470 (2.0140–2.2166)	0.0029
2e-3	**1.8250** (1.8229–1.8283)	**1.8239** (1.8203–1.8324)	0.0095
4e-3	1.8427 (1.7427–2.0220)	2.0378 (1.7692–2.0624)	0.2197
8e-3	2.1271 (2.0509–2.2613)	2.1067 (2.0571–2.1159)	0.1454

- **The best learning rate is the same: 2e-3**, at 1.824 against 1.825.
- **Up to the optimum, all 12 pairs agree within 0.0095.** At these learning rates the seed-to-seed spread is often wider than that (6e-4: 0.15 in ml-ladder's own runs). So pairing by seed is doing real work: it takes the seed noise out of the comparison.
- **Above the optimum, pairs diverge** by up to 0.22 bits/byte. Yet our ranges overlap ml-ladder's, and its own seed spread there is 0.28 (4e-3) and 0.21 (8e-3). The largest gap, seed 2 at 4e-3, is a fork. Both runs are together through step 750 (2.153 against 2.137). Then ours stalls near 1.67 from step 1,250 to 1,750, while ml-ladder's falls to 1.45.

  A control traced this fork to PyTorch's own run, not to JAX or vmap.

seed 2, lr 4e-3, 3,000 steps	bits/byte
ml-ladder's recorded PyTorch run (2026-10-08)	1.8427
ml-ladder's script, rerun twice (same torch 2.11.0+cu128, T4)	2.0121, 2.0413
JAX, sequential (R2's runner), twice	2.0545, 2.0479
JAX, vmapped group of 3, twice	2.0624, 2.0485

  - **All six fresh runs land together,** PyTorch and JAX, sequential and vmapped (2.01–2.06). The recorded 1.8427 is the outlier: PyTorch took the other branch once and did not reproduce it.
  - **Neither framework's T4 runs are deterministic,** so above the optimum one run per (lr, seed) samples a distribution. Its spread at 4e-3 is at least 0.22 bits/byte.

vmap does not make this sweep faster.

copies K	4M, total tok/s (no remat)	4M, with remat	width-128 transformer	width-64 transformer
1	132.5k (1.00×)	103.2k (1.00×)	385k (1.00×)	436k (1.00×)
2	130.9k (0.99×)	104.4k (1.01×)	370k (0.96×)	456k (1.04×)
4	132.4k (1.00×)	106.1k (1.03×)	402k (1.04×)	473k (1.08×)
8	out of memory	107.6k (1.04×)	400k (1.04×)	484k (1.11×)

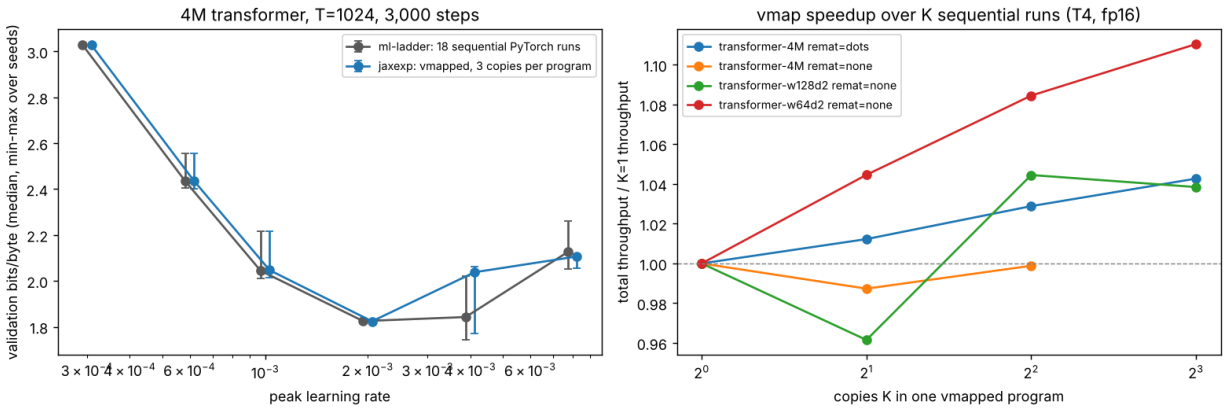


Figure 1: Left: the learning-rate sweep, ours (vmapped) against ml-ladder's sequential runs. Right: throughput of K copies relative to one

- **A single copy already saturates the T4** at 16 × 1024 tokens per step. So K copies take K times as long, and total throughput is flat in K.
- **It stays nearly flat even for a 0.2M-parameter transformer** (1.11× at K=8). With T=1024 and 16 sequences per step, each step's attention and matmuls are large enough to fill the GPU whatever the model width.
- **Memory caps K before speed matters.** XLA's temp memory is 2.87 GiB per copy without remat, so K=4 fits (12.0 GiB) and K=6 does not (15.0 GiB, against 13.8 GiB available). Per-block remat cuts it to 1.26 GiB per copy, so K=8 fits, at 0.78× the speed.
- **The whole sweep cost 1.97 GPU-hours** (six programs of 1,137–1,233 s), against ml-ladder's 1.03 (18 runs × 205 s median). That 0.52× is the same JAX-against-PyTorch gap R2 measured for this model (0.53×), not a cost of vmap.

What vmap *does* buy is that the sweep is one program:

- each group compiles once (9–17 s), not once per run;
- all the copies share one data pipeline;
- the grid is one command and one JSON. N1's sweep tool is now ready for O3 and N2.

What surprised me

PLAN's premise came from the wrong regime. The image probe found that K = 8 copies cost 4.4× one copy, which promised a 1.8× speedup. That probe was a small recurrence whose single copy left the GPU idle. ml-ladder's protocol feeds 16,384 tokens per step, which fills a T4 even with a 0.2M model. The speedup from vmap is the GPU's idle fraction, and here there is almost none.

The attention key bias. Its gradient is exactly zero by symmetry. That showed up as a 5% per-tensor mismatch in what should have been an equality test, and it is invisible in any loss.

The biggest disagreement in the sweep was PyTorch disagreeing with itself. The one pair that looked like a framework difference (0.22 bits/byte) did not survive reruns. ml-ladder's own script, on the same torch version and the same GPU type, landed with JAX (2.01 and 2.04), not at its recorded 1.84. At learning rates above the optimum, runs with identical weights and batches split apart, and nondeterministic GPU arithmetic decides which side a run lands on. The R3 copy task (M2) showed the same thing near its plateau. A single run per configuration in this regime is a draw, not a measurement.

Open threads

- **The small-batch regime.** vmap should pay when one copy cannot fill the device: small batches, short contexts, small models. That is the setting of many hyperparameter studies, and it was not measured here.
- **Deterministic GPU runs** (`--xla_gpu_deterministic_ops`; `torch.use_deterministic_algorithms`), to separate implementation from nondeterminism without repeats.
- **The fork's distribution.** How often does seed 2 at 4e-3 reach about 1.84 rather than about 2.05? Ten repeats would put a number on ml-ladder's luck. The same applies to its 4e-3 entry in §7's table.
- **O3 and N2,** using this sweep tool.

O3 — Optimisers beyond Adam

Question

ml-ladder's B5 thread asks about Lion and Shampoo, and whether second-moment scaling is the right primitive at all. On R2's 4M transformer, with every optimiser at its own tuned learning rate, does anything beat AdamW?

The idea

Three answers to "how should a step be shaped?"

- **AdamW** divides each coordinate's momentum by a running RMS of its gradient: second-moment scaling.
- **Lion** keeps only the *sign* of an interpolated momentum, so every coordinate moves by exactly the learning rate. It has no second moment at all.
- **Muon** orthogonalises the momentum of each weight matrix. It replaces $M = U\Sigma V^\top$ by roughly $UV^\top$, so every direction in the matrix's row and column spaces moves equally. That is Shampoo's preconditioner $(GG^\top)^{-1/4}\,G\,(G^\top G)^{-1/4}$ without its running accumulation: second-moment scaling of a whole matrix, rather than of each coordinate.

Tune every arm (ml-ladder §7, and CLAUDE.md). An untuned baseline produces a fake winner. Each optimiser gets its own learning-rate sweep on seed 0, extended until the best value is interior. Seeds 1–2 then run at that value.

N1's sweep tool makes this cheap to run: each grid is one or two vmapped programs.

The math

With momentum m_t, gradient g_t and peak LR η (on ml-ladder's schedule), and decoupled weight decay λ:

- **AdamW:** $\theta \mathrel{-}= \eta\,(\hat{m}_t/(\sqrt{\hat{v}_t}+\epsilon) + \lambda\theta)$, with $\beta = (0.9, 0.95)$ and $\lambda = 0.1$ (the protocol).
- **Lion:** $\theta \mathrel{-}= \eta\,(\mathrm{sign}(\beta_1 m_{t-1} + (1-\beta_1)g_t) + \lambda\theta)$, $m_t = \beta_2 m_{t-1} + (1-\beta_2)g_t$, with $\beta = (0.9, 0.99)$. Since $|\text{update}| = \eta$, Lion's good learning rates are 3–10× smaller than AdamW's. So $\lambda = 1.0$, ten times AdamW's, to keep $\eta\lambda$ comparable, as the Lion paper recommends.
- **Muon** on each hidden matrix $W \in \mathbb{R}^{n_{\text{in}} \times n_{\text{out}}}$: $W \mathrel{-}= \eta\,(\sqrt{\max(1, n_{\text{out}}/n_{\text{in}})}\ \mathrm{NS}_5(\tilde{m}_t) + \lambda W)$.
  - $\tilde{m}_t$ is Nesterov momentum with $\beta = 0.95$.

- NS_5 is five steps of the quintic Newton–Schulz iteration that pushes the singular values towards 1.
- $\lambda = 0.1$.

Embeddings, the output head, biases and LayerNorm stay on the protocol's AdamW, at AdamW's tuned peak, on the same schedule.

What we measure and why

- **Setting:** the 4M transformer (learned positions) at T=256, batch 64 (16,384 tokens per step), 3,000 steps, fp16 with fused attention. Every run starts from ml-ladder's initial weights.
- **Anchor:** the AdamW arm at 2e-3, 4e-3 and 8e-3 must match ml-ladder's own runs at this setting (`c-gpu-lr2-256.json`).
- **Per optimiser:**
  - the LR curve on seed 0;
  - bits/byte at the best LR over seeds 0–2, as median (min–max);
  - steps to reach AdamW's training loss, at two levels: its level at step 1,500, and its final level. The loss is smoothed as the mean of 8 consecutive logs, 200 steps. With a fixed-length cosine schedule, every run reaches the final level only near the end, so the halfway level is the informative one.

What I built

- **`jaxexp.train.optimizer(kind=adamw|lion|muon).`** The AdamW path is unchanged: R2's parity tests pass.
- **Muon is assembled from `optax.contrib.scale_by_muon` and the protocol's AdamW, via `optax.partition`.** optax's ready-made `muon(nesterov=True)` also applies Nesterov to its AdamW branch, which makes that branch NAdam, with first steps 1.47× larger. A test caught it.

  The AdamW part has its own peak. It follows the schedule through an injected `schedule` multiplier, so one vmapped program can sweep Muon's LR while the AdamW part holds still.
- **`tests/test_o3_optimizers.py`:**
  - Lion's first step is exactly $\pm\eta$ per element;
  - Muon orthogonalises exactly the eight block matrices (singular values in the Newton–Schulz band);
  - everything else gets AdamW's exact first step;
  - both optimisers give the same losses vmapped as sequentially.
- **`n1_vmap_sweep/run.py`** gains `--optimizer`, `--weight-decay` and `--adam-peak`; `experiments/o3_optimizers/analyze.py` scores the results.

Results

The AdamW arm lands on ml-ladder's runs at this setting (seed 0): 1.7141 against 1.7155 at 2e-3, 1.6462 against 1.6436 at 4e-3, and 1.6357 against 1.6239 at 8e-3. ml-ladder's grid stopped at 8e-3, its best value. Extending it shows that 8e-3 really is the optimum: 1.6e-2 gives 1.6912, and 3.2e-2 gives 1.7475.

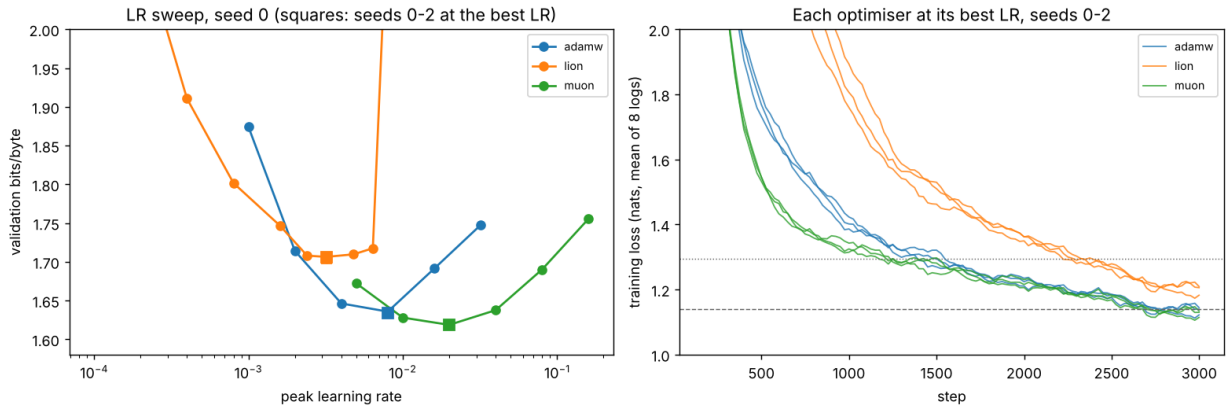


Figure 2: Left: each optimiser's LR sweep on seed 0, with seeds 0-2 at its best LR (squares). Right: smoothed training loss at each optimiser's best LR, seeds 0-2; dotted and dashed lines are AdamW's halfway and final levels

optimiser (tuned)	peak LR	weight decay	bits/byte, seeds 0–2	steps to AdamW's halfway loss	tokens/s (T4)
AdamW (protocol)	8e-3	0.1	1.6335 (1.6333–1.6357)	1,400–1,550	240k
Muon (+ AdamW at 8e-3)	2e-2	0.1	**1.6190** (1.6186–1.6212)	**1,175–1,250**	172k (0.72×)
Lion	3.2e-3	1.0	1.7056 (1.6976–1.7063)	2,225–2,350	234k

- **Muon is best, by 0.0145 bits/byte.** Its three seeds (1.6186–1.6212) and AdamW's (1.6333–1.6357) do not overlap. It reaches AdamW's halfway training loss about 1.22× sooner in steps (median 1,225 against 1,500).
- **But Muon costs more per step.** Five Newton–Schulz iterations on every matrix make it run at 0.72× AdamW's throughput on the T4. Measured in time, it reaches the halfway loss about 14% *later*: 1,225 steps at 1.40× the cost per step is worth about 1,710 AdamW steps. Whether AdamW given 40% more steps would close the 0.0145 gap was not measured.
- **Lion is clearly worse here, after tuning both of its knobs.**
  - Its LR curve is flat around its optimum: 1.7079, 1.7063 and 1.7096 at 2.4e-3, 3.2e-3 and 4.8e-3. It diverges from 1.28e-2.
  - Weight decay matters, and 1.0 is an interior optimum. At 3.2e-3, wd 0.1 gives 3.68 (unstable), 0.3 gives 2.07, and 1.0 gives 1.71; at 1.6e-3, wd 3.0 gives 1.79.
  - The gap to AdamW is 0.072 bits/byte, and Lion needs about 1.5× as many steps to reach AdamW's halfway loss.
- **fp16 was healthy at every tuned setting:** no skipped steps. Steps were skipped only in runs that diverged or nearly did (Lion at 1.28e-2 and above, or with wd 0.1).
- **Cost.** O3 took four kernels: 36 runs of 3,000 steps, as 14 vmapped programs of 2–3 copies, totalling 2.33 GPU-hours of program time on T4s.

What surprised me

Lion needs its weight decay. Lion's step has a fixed size, so without enough decay the weights grow until the sign updates are relatively too small to steer them. Lowering λ from 1.0 to 0.3 cost 0.37 bits/byte at 3.2e-3, and 0.1 made the run unstable. The Lion paper's rule of thumb (λ 3–10× AdamW's) is not a nicety; here it is most of the tuning.

Lion loses at this batch size. Lion's own paper reports its gains at large batches. Here a step is 64 sequences of 256 bytes, and the sign of a noisy momentum throws away the magnitude information that AdamW keeps. That is a hypothesis, not a measurement; the batch-size axis is N2's.

optax's Muon is not plain Muon. `optax.contrib.muon(nesterov=True)` also turns the AdamW it applies to non-matrix parameters into NAdam: a 1.47× larger first step, and a different optimiser for embeddings and the head. It only showed up because the test computed AdamW's first step by hand. The first version of that test failed on gradient clipping and on Adam's ε before it was exact.

A step-count win can be a wall-clock loss. Muon's 1.22× fewer steps become a 14% slower time on a T4, where five Newton–Schulz iterations per matrix are not hidden behind anything.

Open threads

- **Equal-time comparison.** Give AdamW 1.4× the steps, so it has Muon's wall-clock time, and see whether Muon's 0.0145 survives.
- **Muon's details:**
  - orthogonalise q, k and v separately rather than the fused 256×768 matrix;
  - tune its AdamW part's LR jointly;
  - try fewer Newton–Schulz steps, or bf16 iterations on hardware with fast bf16.
- **Lion at larger batches.** Use N2's critical batch size to choose where to look.
- **Longer runs and the 25M model.** 3,000 steps of a 4M model is a short horizon; the ranking may not hold at R2's 12,000 steps.
- **A Shampoo-family optimiser proper** (`distributed_shampoo`, SOAP or PSGD-Kron), once one runs with optax 0.2.8. `psgd-jax` 0.2.9 imports optax internals that no longer exist.

N2 — Gradient noise scale and the critical batch size

Not started. See `PLAN.md`.